

Interpretable Credit Risk Predictor

OVERVIEW

An end-to-end machine learning pipeline that not only predicts whether a client will default on a loan but also provides a mathematical explanation for *why* the model made that specific decision, addressing the "black box" problem of AI.

Category	Specification
Tech Stack	Python, Pandas, Scikit-Learn, XGBoost, SHAP
Dataset	UCI Default of Credit Card Clients Dataset
Target Metric	Maximize AUC-ROC and F1-Score on the test set
Output	A trained model and visual SHAP plots (Summary and Force plots)

FUNCTIONAL REQUIREMENTS

- **Data Preprocessing:** The pipeline must handle missing values, encode categorical variables (like education or marital status), and scale continuous variables (like credit limit).

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- **Imbalance Handling:** The pipeline must utilize SMOTE (Synthetic Minority Over-sampling Technique) to balance the training data, as default cases are significantly rarer than non-defaults.
 - **Model Training & Tuning:** The system must train an XGBoost Classifier, utilizing Grid Search or Random Search to optimize hyperparameters (`learning_rate`, `max_depth`, `n_estimators`).
 - **Global Interpretability:** The system must generate a SHAP Summary Plot to illustrate which features most heavily influence the model's behavior across the entire dataset.
 - **Local Interpretability:** The system must generate a SHAP Force Plot for a single, randomly selected user, showing exactly how their specific age, balance, and payment history pushed their individual prediction toward or away from default.

MILESTONES FOR COMPLETION

1. Complete Exploratory Data Analysis (EDA) and preprocess the data.
2. Train the XGBoost model and print the classification report.
3. Integrate the SHAP library and export the visual explanation plots.